Pressure Calculations for the pneumatic Gripper:

1. Given Data

Object mass: **m = 200 g = 0.2 kg**

Gravity: **g = 9.81 m/s²**

From your CAD:

Radius ≈ **14.538 mm = 0.0145 m**

2. Calculate Object Weight (Force)

We use:

$$F = m \cdot g$$

So:

$$F = 0.2 \times 9.81 = 1.962 \text{ N}$$

This is the **minimum force needed to just hold the object (no safety)**

3. Apply Safety Factor

For grippers, we **always multiply** because of:

Slipping

Vibrations

Imperfect contact

Typical safety factor = **2 to 4**

Let's use **3**:

$$F_{required} = 1.962 \times 3 \approx 5.9 \text{ N}$$

4. Convert to Gripping Force per Finger

Since your gripper has **2 jaws**:

$$F_{per\ jaw} = \frac{5.9}{2} \approx 2.95 \text{ N}$$

5. Include Friction (VERY IMPORTANT)

Gripping depends on friction:

$$F_{grip} = \frac{F_{load}}{\mu}$$

assume:

Rubber contact → μ ≈ **0.5** (reasonable)

$$F_{grip} = \frac{5.9}{0.5} = 11.8 \text{ N}$$

total gripping force needed = ~12 N

6. Convert Force → Required Pressure
Now the key formula:

$$P = \frac{F}{A}$$

Where:
A = piston area

Assume Cylinder Size
Let's assume a **typical small pneumatic cylinder**:
Diameter = **20 mm → radius = 10 mm = 0.01 m**

$$A = \pi r^2 = \pi(0.01)^2 = 3.14 \times 10^{-4}\, m^2$$

Calculate Pressure

$$P = \frac{12}{3.14 \times 10^{-4}} \approx 38216\, Pa$$
$$P \approx 0.038\, MPa \approx 0.38\, bar$$

To determine the required gripping force, I started by assuming an object mass of 200 grams, which corresponds to a weight of approximately 1.96 Newtons under gravity. Since this value only represents the minimum force needed to hold the object under ideal conditions, I applied a safety factor of 3 to account for uncertainties such as vibrations, misalignment, and surface irregularities. This increased the required force to about 5.9 Newtons. Additionally, I considered the effect of friction between the gripper and the object. Assuming a coefficient of friction of 0.5, the required gripping force was further increased to approximately 12 Newtons to ensure a stable and slip-free grasp.

After determining the required gripping force, I calculated the pneumatic pressure needed to achieve this force using the gripper's geometry. Based on the CAD model, the effective radius of the piston was 14.53 mm, which corresponds to an area of approximately 663 mm^2. Using the relationship between force and pressure, I found that the minimum required pressure is around 0.18 bar. This relatively low pressure is sufficient for handling lightweight objects and offers advantages such as reduced energy consumption and safer operation. However, to ensure reliability in real-world conditions, it is recommended to operate within a slightly higher pressure range, typically between 0.3 and 0.6 bar.